

24MAT110-ASSIGNMENT 1

1. Find the domain and range of the following functions.

(a) $f(x) = 5 - \sqrt{x^2 - 2x - 3}$

(b) $f(x) = 2 + \frac{3x^2}{x^2 + 4}$

2. Find the following limits without using L'Hospital's rule.

(a) $\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x - 1}$

(b) $\lim_{x \rightarrow 1} \frac{x^{2/3} - 1}{1 - x^{1/2}}$

(c) $\lim_{x \rightarrow 0} \frac{|2x + 2| - |x| - 2}{x}$

(d) $\lim_{x \rightarrow 0} \frac{x + x \cos x}{\sin x \cos x}$

(e) $\lim_{x \rightarrow 1} \frac{x^{50} - 1}{x - 1}$

3. Given a function $f(x) = \begin{cases} \sqrt{1 - x^2} & 0 \leq x < 1 \\ 1 & 1 \leq x < 2. \\ 2 & x = 2 \end{cases}$

(a) Draw the graph of the function.

(b) Find the domain and range of the function.

(c) At what points does the limit exist? At what points does it not exist? Explain.

(d) At what points is the function continuous? What different types of discontinuities are there for $f(x)$? Justify.

4. Check the continuity of the function $f(x) = \begin{cases} -3x - 6 & x \leq 0 \\ 2x^2 - 6 & 0 < x \leq 2. \\ 3x - 5 & x > 2 \end{cases}$

5. Show that the function $f(x) = (x - a)^2(x - b)^2 + x$ takes on the value $\frac{a + b}{2}$ for some values of x .

6. If the product function $h(x) = (x) \cdot g(x)$ is continuous at $x = 0$, must (x) and $g(x)$ be continuous at $x = 0$?

7. Check the differentiability of the function $f(x) = \begin{cases} x^{2/3} & x \geq 0 \\ x^{1/3} & x < 0 \end{cases}$ at the origin.

8. Find the two points where the curve $x^2 + xy + y^2 = 7$ crosses the x-axis, and show that the tangents to the curve at these points are parallel.